

1. Dissertation Title + Terminology

Subject: Proposed dissertation title + terminology closure — please review

I've thought this through and propose:

“LayerIt!: Towards a Time-Aligned Framework for Composable Music Visualisation Layers”

1.1 Title, term by term

- **“Towards a”** — signals this is a prototype/feasibility-stage contribution, not a finished, general-purpose framework. Keeps the claim honest.
- **“Time-Aligned”** — describes the *end-state* of the final composition (all layers coherently share a physical-time axis), not a precondition every layer must individually satisfy before it's allowed in.
- **“Framework”** — the core contribution: a system for building compositions, not a single algorithm.
- **“Composable”** — layers are modular, reusable, combinable (your OOP/layers design principle).
- **“Music Visualisation Layers”** — the object being composed.

1.2 Notes

What LayerIt is

- LayerIt is a **visualisation composition framework**. Its job is to let you stack, combine, and reuse layers (audio waveform, tempo curve, score, physiological data, etc.) into one coherent, time-aligned view.
- It is **not** a score aligner. Alignment is a means, not the end.

Where alignment fits

- **Foundational dichotomy:** every layer's native time basis is either **physical time** (elapsed duration) or **abstract time** (metrical/structural position). This distinction drives everything below.
- Some layers are **already physical-time-native** (audio, tempo/dynamics curves, beats/downbeats/onsets positions, probabilities) — no alignment step needed, they slot straight in.
- Some layers are **abstract-time-native** — their intrinsic axis is metrical/structural position, not duration. In this thesis, the only implemented case is the **score**, though the category is general enough to admit others later.
- Only abstract-time layers require a **time-alignment** step to enter the composition. **ScoreWarp is one implementation of that step — one component LayerIt can call, not what LayerIt is.**

How does this connects to the original annotation problem

- This will have to be explored later in discussion/conclusion and future work. Although you haven't completed this part (ie, integrate your solution in an annotation tool), this is a very important first step towards that goal. Later in discussion, we'll have to argue that for an annotator, it is completely different to see a link-based (anchor) alignment (as in Piano Precision) instead of a warped score and several time-aligned layers. It opens many possibilities (e.g. to highlight high-probability beat regions).

1.3 Terminology to cite (use these exact labels when tagging sources in your notes/bib)

- **goebl2025_scorewarp** — Goebel & Weigl, “Let's do the ScoreWarp again!” (MEC 2025).
 - Source of: “physical time,” “warp”/“warped score,” and the related-work category of angled-line indicators connecting score to a physical timeline.
- **weigl2020_trompa** — Weigl, TR-D3.5 Multimodal Music Information Alignment v2 (TROMPA).
 - Source of: “anchor points”/“anchor types” (formal correspondence-tuple vocabulary), and “layers of abstraction” (measures/beats/notes) — the seed for our “abstract time.”
- **jiang2025_pianoprecision** — Jiang, Weigl & Goebel, Piano Precision (SMC 2025).
 - The working example of link-based (anchor) alignment: score keeps its native engraved layout; correspondence is shown via highlighting/connectors rather than by moving anything. Also gives us the field's shared term “alignment” itself (cf. SMAT, MAPS, MATCH tool names).

1.4 Alignment strategy pair (define explicitly in the dissertation – see other comments)

- **Time-warped (continuous) alignment** — reflow the layer's own geometry onto physical time → goebl2025_scorewarp
- **Link-based (anchor) alignment** — preserve native layout, express correspondence via connectors/anchors → jiang2025_pianoprecision, formalised by weigl2020_trompa